* is accordance to principle of transmissibility. point B

Proof

* Provide as equal and oppos forces at B such that force in A and a force in B ~~from~~ form couple with d as distance b/w them.
* Thus the force in A shift to B and a couple is formed at B.
* A given force F applied to a point A can be moved along its line of action but cannot be moved to any point away from the line of action without modifying the action of F on the rigid body.
* Consider a point B which is at a dis d from the line of action of force F, 2 oppositely directed collinear forces each of magnitude f and line of action parallel to the line of action of given force F at a are introduced at B.
* The force F at A and oppositedly directed force at B consitute a couple of moment i.e, $M = f \times d$
* Thus the force F acting at A can be resolved into a force acting at B together with a couple of magnitude $f \times d$.

~~Book~~

Benjamin Sir Tb
(Engg Mechanics)

* Attendance : 10 marks
* Assignment quiz - 15 marks
* Continuous Assesment test - 25 marks

force system - Several force of different magnitude acting on a body

Coplanar
* All the force in same plane

Non-coplanar
* force act in different plane

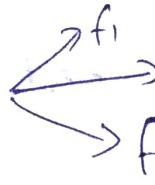
Coplanar: collinear, parallel, concurrent, non-concurrent

Non-coplanar: parallel, concurrent, non-concurrent

i) coplanar collinear : forces on same line , eg. tug of war

ii) coplanar parallel : Same line and parallel line of motion eg: system of wires load acting on a beam

iii) coplanar concurrent : forces on same plane and through same point.

eg: 

force on a rode resting against a wall

iv) Coplanar non-concurrent : forces on same plane and through different point

v) non-coplanar parallel : forces on diff plane having parallel line of action

vi) non-coplanar non-concurrent : forces on diff plane and diff point of action

vii) non-coplanar concurrent :